Regression and Random Variables

Chapter 3. Least squares estimation

The normal equations for a simple linear regression of y on x are written below.

$$y = a + b x + e$$

Coefficient estimates

x	y
1	2.5
2	4.1
3	5.8

The fitted line and the residuals

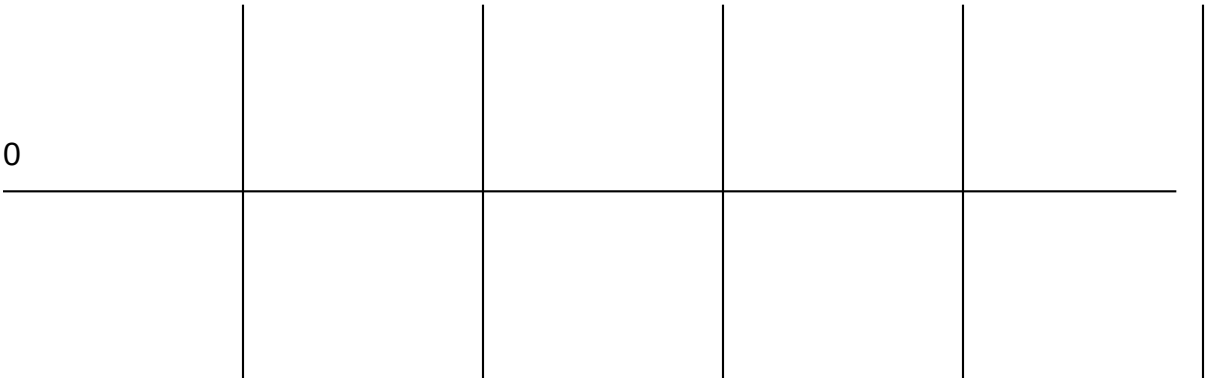

Table 3.1 gives the same coefficients.

Implementation notes

A short listing that fits the dataset follows.

The slope is the estimate of the effect.

The intercept is the level at x = 0.